Road Safety Intervention Targeting

Client Project Status Report

Project: STATS19 Road Safety Analysis
Team: Team 1 | Reporting period: 23 Feb – 5 Mar 2026
Date produced: 05 March 2026

EXECUTIVE OVERVIEW AND PROGRESS

Overall Status	● ON TRACK

Executive Summary

- **Achieved this period:** Project scoped and governance established, with the full STATS19 dataset (1979–2024, 1.9M collision records) downloaded, explored, and filtered from 2015–2024. We completed the initial severity and road user analysis with four client-ready figures produced.
- **Early signal:** Fatal collisions are constant at approximately 1,500–1,700 per year (2020–2024), while serious collisions have risen from ~18,000 (2020) to ~21,000 (2024); car occupants, pedestrians, and motorcycle riders account for the majority of fatal and serious casualties, however this serious trend must be treated with caution given the post-2016 severity reclassification.
- **Next period:** Data cleaning, hotspot analysis, and the first full reproducible pipeline will be completed; no client input is currently required to maintain momentum.

Key Milestones Achieved

● Scope agreed and SELECTION.md uploaded	23 Feb 2026
● Repository and workspace established	23 Feb 2026
● Management plan finalised and version-controlled	23 Feb 2026
● Full STATS19 dataset downloaded and explored	28 Feb 2026
● Initial severity and road user figures produced	05 Mar 2026

Top 3 Risks / Issues

1. Risk / Issue	**Data join limitation — full STATS19 collision file incomplete**
Why it matters	The full 1979-present collision CSV contains only ~21% of expected records, so trying to join all three tables across that entire timeframe just isn't going to work.
Mitigation / Owner	Using 5-year collision file (2020–2024) for geographic analysis; 10-year casualty file for trend analysis. Scope split documented. Owner: PA.
Next update	12 Mar 2026
2. Risk / Issue	**Severity reclassification bias (post-2016)**
Why it matters	The 2016 injury-based reporting change inflates Serious counts, risking overstated trend conclusions if raw severity is used.
Mitigation / Owner	Adjusted severity columns confirmed 0% missing from 2015+. All analysis will use adjusted figures. Owner: PA.
Next update	12 Mar 2026
3. Risk / Issue	**Exam period compressing final reporting weeks**
Why it matters	Team availability reduces in late April, risking rushed QA and writing.

Mitigation / Owner	Internal completion target set to 20 April (one week buffer). Executive summary skeleton to begin week of 16 March. Owner: All.
Next update	20 Apr 2026

DETAILED PERFORMANCE AND NEXT STEPS

Work Accomplished This Period

Local auhtority refrence dataset integrated

ONS local authority district lookup table imported to support geographic aggregation and mapping of collision records to administrative areas. Dataset contains 422 authorities with standard ONS codes and names, enabling consistent spatial analysis and future joins with external datasets.

Evidence: local-authority-ons-district-names.csv

Data integration strategy defined

Collision, vehicle, and casualty tables assessed for relational linkage. Structural differences in join keys across STATS19 years identified and mitigation strategy defined prior to modelling.

Evidence: data_join_notes.md, join_validation_summary.csv

Project governance established

Scope agreed, GitHub repository created, management plan with risk register and QA plan finalised.

Evidence: management_plan.md, action_log.md, contact_log.md

Full STATS19 data acquired and explored

Three tables loaded (1.9M collisions, 3.7M vehicles, 6.9M casualties). Column structure, missingness, and join key behaviour documented across all tables.

Evidence: outputs/tab/missingness_summary.csv

2015–2024 filter applied and validated

All three tables filtered to 2015 onwards. Latitude/longitude and adjusted severity confirmed 0% missing post-filter. Join key structural issue identified and mitigation agreed. *Evidence: outputs/tab/filter_summary.csv*

Initial severity and trend analysis completed

Fatal collisions flat ~1,500–1,700/yr; Serious rising from ~18k (2020) to ~21k (2024). Car occupants dominate volume; pedestrians and motorcyclists disproportionately at risk.

Evidence: fig1_yearly_trend.png, fig2_severity_breakdown.png, fig4_casualty_type.png

COVID robustness check completed

Excluding 2020–2021 does not change trend direction. Result is not fragile.

Evidence: fig3_covid_robustness.png, table3_covid_robustness.csv

Generalized Linear Model (GLM) anaysis implemented

Poisson log-link GLM applied to model daily road casualties over time. Interrupted time series specification used to test the impact of the 23 March 2020 COVID-19 lockdown. Results sho break associated with the policy intervention. a statistically significant immediate reduction in casualties (rate ratio ≈0.95) and a subsequent change in trend, indicating a structural

Evidence: glm_results_table.csv, glm_lockdown_model.png, glm_time_series_plot.png

Planned Work — Next Two Weeks

Planned Item / Work Package	Owner	Target Date	Output Expected
Full data cleaning pipeline (recode -1, deduplication, validation)	ZA	13 Mar	clean_joined.csv, exclusions_log.csv
Hotspot analysis — geographic severity mapping by local authority	PA	23 Mar	fig_hotspot_map.png, table_hotspot.csv
Contributory factor analysis — speed limit, road type, junction, light	PA	23 Mar	fig_factors.png, table_factors.csv
Alternative severity filter robustness check (post-2016 only)	PA	23 Mar	table_robustness_severity.csv
Executive summary skeleton drafted	CS	16 Mar	report/executive_summary_v0.md
First full reproducibility clean run from repo root	AS	13 Mar	QA log entry, reproducibility_record.md

Decisions Required

Decision 1: Confirmation of agreed analysis time window - **By:** 10 Mar 2026 — **Why it matters:** Determines whether trend analysis runs 2015–2024 (casualties only) or 2020–2024 (full three-table join). Delay risks duplicated analysis work. It basically comes down to a trade-off: do we want to look at long-term trends or focus on the more accurate, recent joins? Can change???????

Decision 2: Client confirmation on geographic resolution required for hotspot output (local authority vs. police force area) — **By:** 10 Mar 2026 — **Why it matters:** Affects hotspot definition and the robustness check design. Without this, both alternatives must be run in parallel at additional cost.

Decision 3: Confirmation that sensitive contributory factors data request has been submitted to DfT (roadacc.stats@dft.gov.uk) — **By:** 12 Mar 2026 — **Why it matters:** Contributory factors data requires a separate DfT application. If not submitted this week, the factor analysis will be limited to non-sensitive fields only.

Decision 4: Policy Intervention Data for "Before/After" Analysis. To meet the objective of assessing recent policy impact , the team requires a specific list of implementation dates for local 20mph rollouts and school-street measures.+2 — **By:** 12 Mar 2026 — **Why it matters:** Without these dates, we cannot provide the requested "before/after" evidence-informed view and will be limited to purely descriptive trend analysis.

Decision 5: Threshold for "Material Change" Approval. We have drafted a working definition of "materiality" as a sustained 10-20% change over 2-3 years for KSI. We need client sign-off on whether this meets their operational needs for a "practical definition".+2 — **By:** 12 Mar 2026 — **Why it matters:** Risk of producing hotspot rankings that do not align with the client’s operational capacity or definition of urgency. These are some decisions Gemini gave me